



22MAT220 - MATHEMATICS FOR COMPUTING 3

Analog Computing Project Report

Team C7

Topic : Simple Harmonic Motion

Team Members

- **G Prajwal Priyadarshan - CB.SC.U4AIE24214**
- **Kabilan K- CB.SC.U4AIE24224**
- **Kishore B - CB.SC.U4AIE24227**
- **Rahul L S - CB.SC.U4AIE24248**

OBJECTIVE

The goal of this project was to understand the fundamental principles of analog computing by building and testing an electronic circuit that solves a second-order linear differential equation. The chosen equation was for Simple Harmonic Motion, a classic example that forms the basis of all oscillators.

ADALM2000 OVERVIEW

The ADALM2000, also known as the Active Learning Module M2K, is a versatile, USB-powered device for data acquisition and signal processing, crafted by Analog Devices. Tailored for educators, makers, and professionals, it enables practical exploration of analog and digital circuits. Serving as a mobile workbench, the M2K consolidates tools like oscilloscopes, signal generators, and voltage sources into one unit, perfect for interactive circuit design and validation using the intuitive Scopy application.

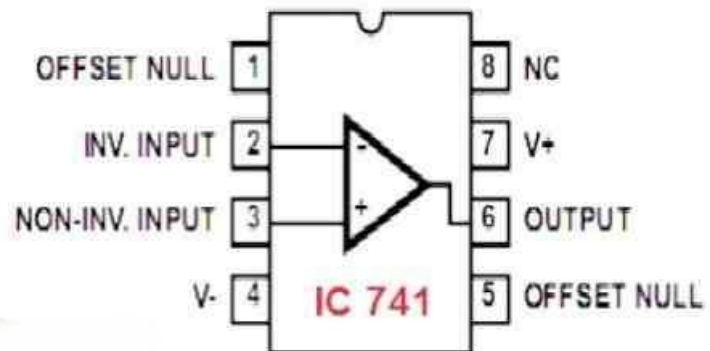


Through its 12-pin interface, the ADALM2000 delivers a range of capabilities: dual analog inputs for scoping signals, dual analog outputs for waveform creation, GPIO lines, regulated power options ($\pm 5V$ and $+3.3V$), plus a common ground. This setup empowers real-time signal monitoring, synthesis, and manipulation for both analog and digital domains. Its compact design and broad utility make it a staple in academic settings and rapid prototyping.

setups, facilitating circuit evaluation, embedded system troubleshooting, and dynamic signal analysis.

THE OPERATIONAL AMPLIFIER (OP-AMP)

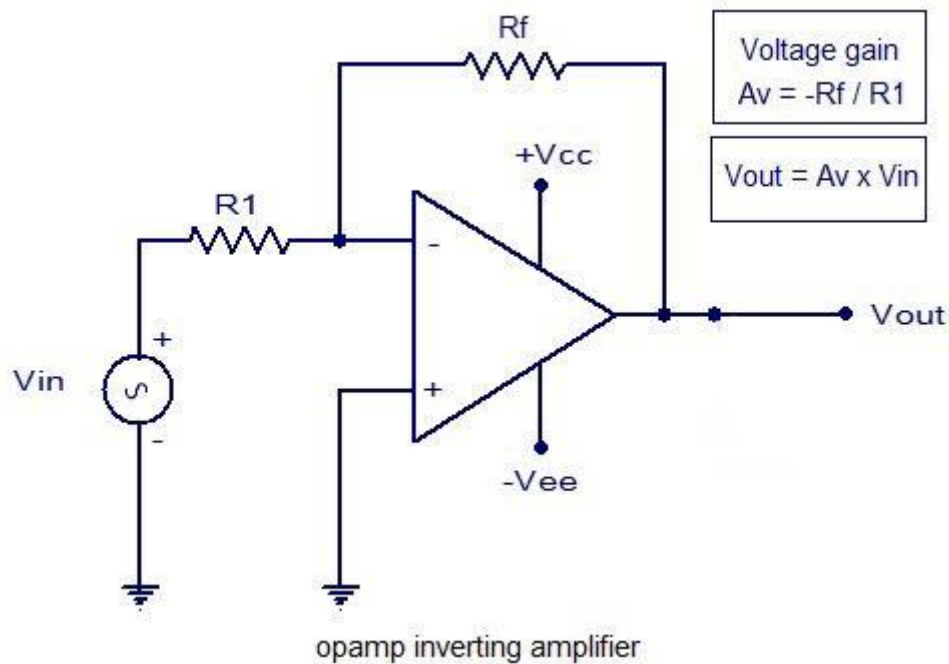
The μ A741, or 741 op-amp, is a versatile IC for signal amplification, filtering, and analog processing. It offers high voltage gain, high input impedance, and low output impedance, ideal for amplifiers and signal conditioners. Internally compensated for stability, it operates on a $\pm 15\text{V}$ dual supply.



Packaged in an 8-pin DIP, its pinout includes: pins 1 and 5 for offset null; pin 2 (inverting input); pin 3 (non-inverting input); pin 4 (V^-); pin 6 (output); pin 7 (V^+); pin 8 (unused). Valued for reliability and simplicity, it's commonly used in amplifiers, filters, oscillators, and comparators for education and prototyping.

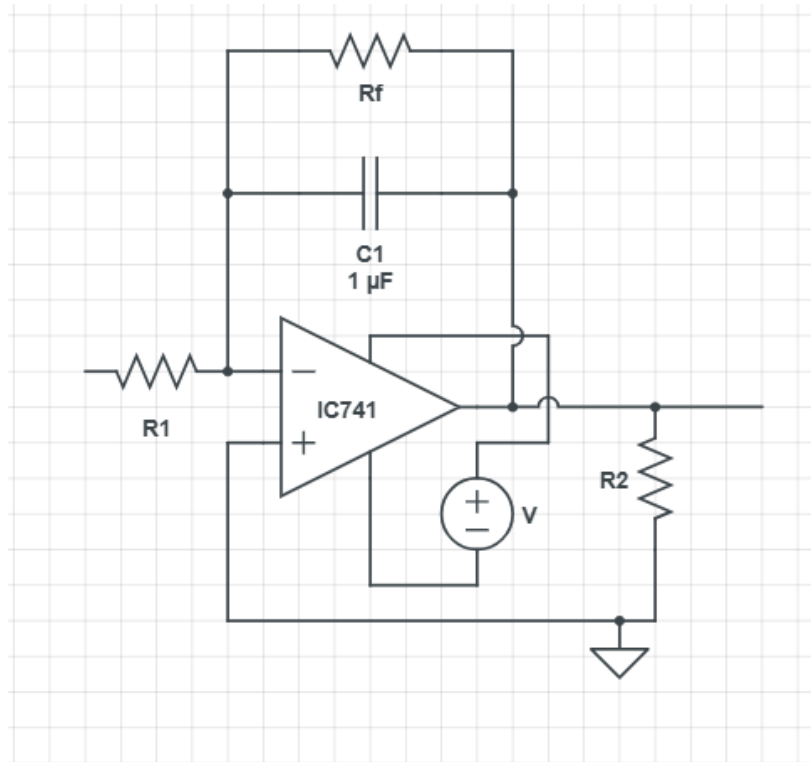
INVERTING AMPLIFIER

- An inverting amplifier uses a standard op-amp arrangement, feeding the input signal via a resistor to the inverting ($-$) input, with the non-inverting ($+$) input tied to ground. This results in an output that's phase-shifted by 180° , effectively flipping the signal's polarity. Gain is set by the proportion of feedback and input resistors, ensuring consistent and accurate amplification.
- Here, the input resistor (R_1) links the signal to the inverting pin, while the feedback resistor (R_2) loops from output to inverting input. The gain formula is $A_v = -(R_2 / R_1)$, with the minus sign showing inversion. Thus, $V_{out} = -(R_2 / R_1) \times V_{in}$. It's a staple in audio blending, signal manipulation, and automation circuits needing reliable inverted boosting.

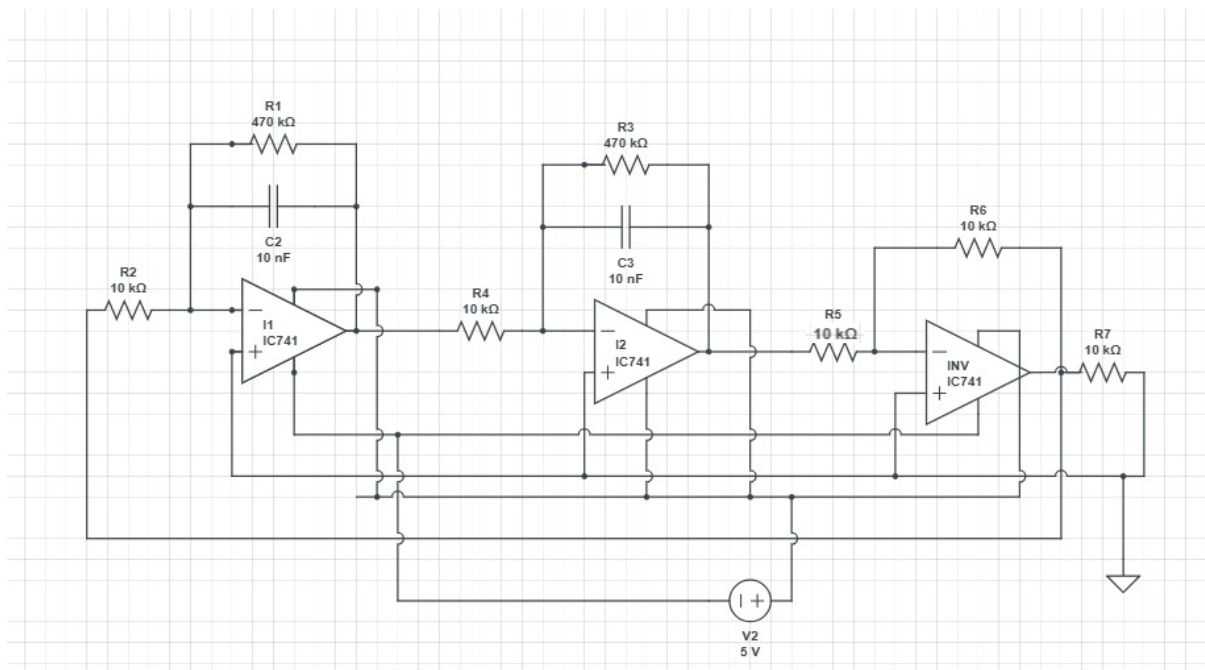


INTEGRATOR

- An op-amp integrator employs a classic operational amplifier setup, routing the input signal through a resistor to the inverting (–) terminal, while grounding the non-inverting (+) input. This configuration yields an output voltage that accumulates the input's value over time, performing mathematical integration and often introducing a 180° phase shift for polarity reversal.
- In this design, the input resistor (R) connects the signal to the inverting node, with a capacitor (C) in the feedback path from output to inverting input. The integration relationship is $V_{out} = -(1/(RC)) \int V_{in} dt$, where the integral represents the time-based summation. It's essential in waveform synthesis, control systems, and analog computation tasks requiring smooth accumulation of signals.



SIMPLE HARMONIC OSCILLATOR CIRCUIT



SHM Differential Equation

1. The governing equation for Simple Harmonic Motion is:

$$\frac{d^2x}{dt^2} + \omega_0^2 x = 0$$

2. Circuit Stage Equations

We assume ideal op-amps and V_{in} use as the voltage feeding the first integrator, V_1 as the first integrator output, and V_2 as the second integrator output.

Stage 1: First Integrator

With input V_{in} and components R_1, C_1 :

$$V_1(t) = -\frac{1}{R_1 C_1} \int V_{in}(t) dt$$

Differentiating to get the relationship:

$$\frac{dV_1}{dt} = -\frac{1}{R_1 C_1} V_{in} \quad (\text{Eq . 1})$$

Stage 2: Second Integrator

With input V_1 and components R_2, C_2

$$V_2(t) = -\frac{1}{R_2 C_2} \int V_1(t) dt$$

Differentiating to get the relationship:

$$\frac{dV_2}{dt} = -\frac{1}{R_2 C_2} V_1 \quad (\text{Eq . 2})$$

Stage 3: Inverting Amplifier (Unity Gain)

For sustained oscillation, a unity-gain inverting amplifier is typically used connecting V_2

back to the first integrator's input V_{in} :

$$V_{in} = -\frac{R_f}{R_3} V_2$$

$$V_{in} = -V_2 \quad (\text{Eq. 3})$$

3. Deriving the Final Differential Equation

Step A: Substitute (Eq. 3) into (Eq. 1)

Substitute $V_{in} = -V_2$ into the first integrator equation

$$\frac{dV_1}{dt} = -\frac{1}{R_1 C_1} (-V_2)$$

$$\frac{dV_1}{dt} = \frac{1}{R_1 C_1} V_2 \quad (\text{Eq. 4})$$

Step B: Differentiate (Eq. 4)

$$\frac{d^2 V_1}{dt^2} = \frac{1}{R_1 C_1} \frac{dV_2}{dt} \quad (\text{Eq. 5})$$

Step C: Substitute (Eq. 2) into (Eq. 5)

$$\frac{d^2 V_1}{dt^2} = \frac{1}{R_1 C_1} \left(-\frac{1}{R_2 C_2} V_1 \right)$$

Step D: Rearrange to SHM Form

$$\frac{d^2 V_1}{dt^2} + \left(\frac{1}{R_1 C_1 R_2 C_2} \right) V_1 = 0 \quad (\text{Eq. 6})$$

4. Frequency Equations

By comparing (Eq. 6) with the general SHM equation $\frac{d^2 x}{dt^2} + \omega_0^2 x = 0$

$$\omega_0^2 = \frac{1}{R_1 C_1 R_2 C_2}$$

$$\omega_0 = \frac{1}{\sqrt{R_1 C_1 R_2 C_2}} \quad (\text{rad/s})$$

Oscillation Frequency (Hertz)

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{R_1C_1R_2C_2}} \quad (\text{Hz})$$

PROJECT CIRCUIT: THE SIMPLE HARMONIC OSCILLATOR

The specific equation we solved is:

$$\frac{d^2x}{dt^2} + \omega_0^2x = 0$$

Here ,

$$\omega_0 = 10^4$$

$$f = 1.6 \text{ khz}$$

This means "acceleration is the opposite of position." To build this, we created a three-stage loop that continuously solves this equation.

Bill of Materials

- 3 x 741 Op-Amp ICs
- 4 x 10 kΩ Resistors
- 2 x 0.1 μF Capacitors
- Breadboard and Jumper Wires

Complete Assembly and Connections

This is the final, step-by-step guide for the complete circuit.

Power Connections

- Connect ADALM2000 **V+** and **V-** to the breadboard power rails. Create a **ground rail**.
- For **all three ICs**, connect **Pin 7** to the **V+** rail and **Pin 4** to the **V-** rail.

Stage Connections

- **IC #1 (Inverter):**
 - **Pin 2 -> Feedback resistor to Pin 6 and Input Resistor**
 - **Pin 3 -> Ground**
 - **Pin 4 -> V(-)**
 - **Pin 6 -> connects to the Pin 2 of IC #2.**
 - **Pin 7 -> V(+)**
- **IC #2 (First Integrator):**
 - **Pin 2 -> Feedback resistor, capacitor to Pin 6 and Input Resistor**
 - **Pin 3 -> Ground**
 - **Pin 4 -> V(-)**

- Pin 6 -> connects to the Pin 2 of IC #3 and to Ground.
- Pin 7 -> V(+)
- IC #3 (Second Integrator):
 - Pin 2 -> Feedback resistor, capacitor to Pin 6 and Input Resistor
 - Pin 3 -> Ground
 - Pin 4 -> V(-)
 - Pin 6 -> connects to the Pin 2 of IC #1 and to Ground.
 - Pin 7 -> V(+)

Loop Connections

1. **Output of IC #3 (Pin 6)** → connects to the input resistor of **IC #1**.
2. **Output of IC #1 (Pin 6)** → connects to the input resistor of **IC #2**.
3. **Output of IC #2 (Pin 6)** → connects to the input resistor of **IC #3**.

Testing and Final Result

- **Test Connections:**
 - The **Signal Generator (W1)** is not used.
 - Oscilloscope **1+** connects to the **Output of IC #3 (Pin 6)** to see position (x).
 - Oscilloscope **2+** connects to the **Output of IC #2 (Pin 6)** to see velocity ($-\dot{x}$).
 - Both **1-** and **2-** connect to the **ground rail**.

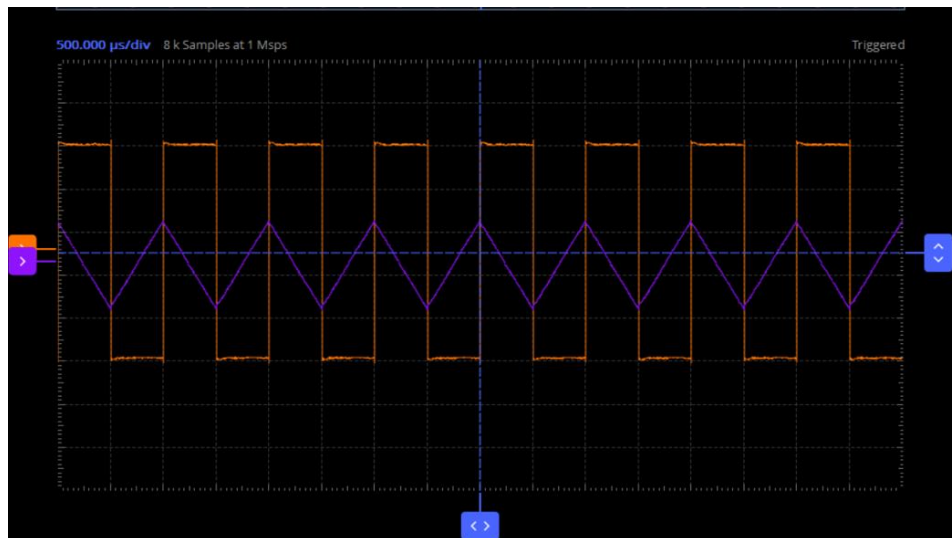
6.

Final Computation:

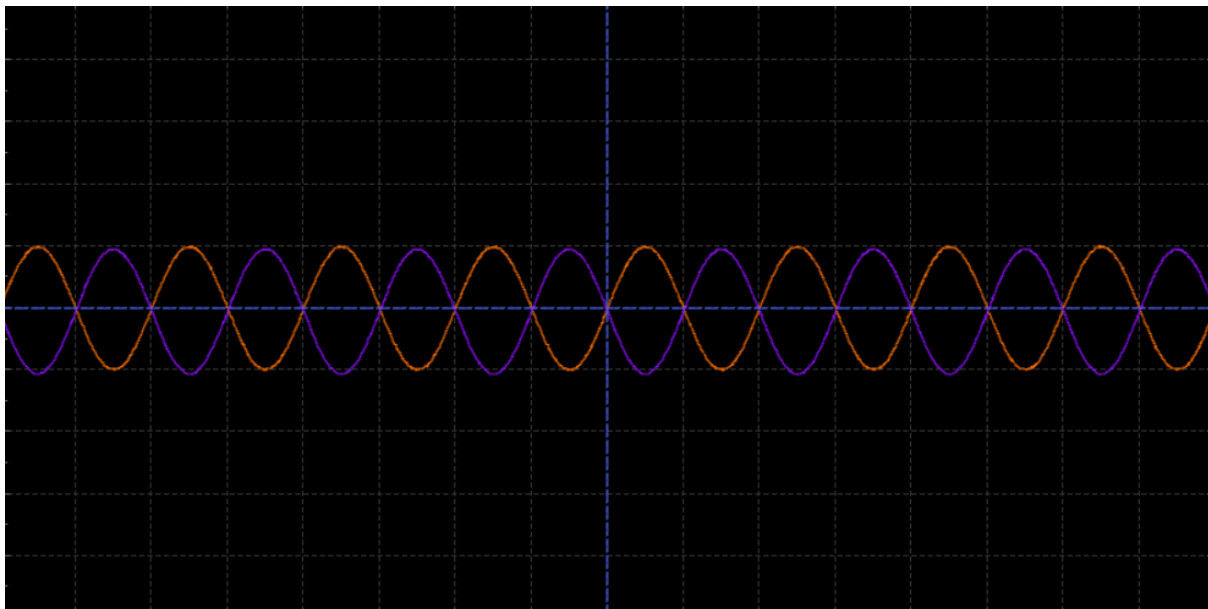
- After powering the initial condition, wait for 5 seconds and remove the wire.
- Go to oscilloscope to get the output of the SHM with a as the Channel-1 (Orange line) and v in the channel 2 (Violet line).
- The final Expected output is to be sinusoidal wave in orange, and output violet line should be phased by 90° .

Output :

- **Integrator output to demonstrate the working of Integrator**



- Inverting amplifier Output



Expected Output :

Based on the output in Simulink model

When we connect integrator 1 and integrator 2 output to Scope

